KLH

Department of Electronics and Communication

EXPLORATORY DATA ANALYSIS REPORT

Hybrid GNN-Based NOMA for User Assignment and Slot Pairing

Project Report

Submitted in partial fulfillment of the requirements
for the Final Year Project

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Abstract

This report presents an exploratory data analysis (EDA) of the dataset prepared for the project *Hybrid GNN-Based NOMA for User Assignment and Slot Pairing*. The project investigates graph-based learning for user assignment and pairing decisions in a Non-Orthogonal Multiple Access (NOMA) scenario.

The dataset contains 290 scenarios, 2,900 user records, and 26,100 directed edge records. Each scenario contains 10 users and considers five slots. The data include user-level features, graph connectivity information, edge-level features, oracle pairing labels, and scenario-level performance summaries.

The analysis examines the dataset structure, split organization, feature consistency, label distribution, correlations, data quality, and limitations relevant to model development. The dataset is a synthetic reconstruction designed to reproduce the configured scenario distributions and labeling logic. It should therefore not be interpreted as a collection of real-world wireless measurements or as an exact reproduction of numerical samples from an earlier simulator run.

Keywords: Exploratory Data Analysis, Graph Neural Networks, NOMA, User Assignment, Slot Pairing, Synthetic Dataset.

1. Introduction

1.1 Project Background

Non-Orthogonal Multiple Access (NOMA) enables multiple users to share communication resources through non-orthogonal transmission and suitable interference management. User assignment and slot pairing can influence achievable rate, interference, and outage performance.

Graph Neural Networks (GNNs) provide a framework for learning from relationships between entities. In this project, users can be represented as graph nodes and relationships between users as graph edges. Node and edge features can then be used to learn or evaluate user-assignment and pairing decisions.

1.2 Purpose of the EDA

The purpose of this report is to understand the dataset before training or evaluating a learning model. The analysis aims to:

- Examine the number of scenarios, users, and edges.
- Understand the organization of the dataset splits.
- Identify the roles of node-level and edge-level features.
- Inspect oracle pairing labels and class imbalance.
- Identify constant, redundant, or potentially uninformative features.
- Check basic data quality and consistency.
- Document limitations that may affect model evaluation.

1.3 Scope

This report focuses on structural and descriptive analysis of the supplied dataset. It does not claim that a trained GNN has achieved any particular accuracy, throughput

improvement, or generalization performance. Such conclusions require model training and a separate evaluation procedure.

2. Dataset Description

2.1 Dataset Overview

The supplied dataset is organized around scenario-based graph instances. Each scenario contains a group of users and directed edges representing relationships between users.

Table 2.1: Overall dataset summary

Item	Count or Description
Total scenarios	290
Total user records	2,900
Total directed edge records	26,100
Users per scenario	10
Number of slots	5
User feature columns	36
Edge feature columns	10
Scenario summary columns	10

The user table contains node-level records, the edge table contains directed relationships, and the scenario summary contains aggregated performance information.

2.2 Dataset Splits

The dataset provides training, validation, and benchmark partitions.

Table 2.2: Dataset split organization

Split	Scenarios	Users
Training	200	2,000
Validation	40	400
Benchmark	50	500
Total	290	2,900

The training split is intended for model development. The validation split can be used

for model selection and tuning. The benchmark split is intended for simulator-oriented evaluation and should be used according to the project's evaluation protocol.

Important: All users and edges belonging to a scenario should remain in the same split. Randomly dividing individual user or edge rows across splits could introduce information leakage because records from one scenario are related.

2.3 Scenario and Graph Structure

Each scenario contains 10 users and considers five slots. The graph representation includes user-level information and directed edge-level relationships.

For a scenario graph, node features describe individual users, while edge features describe relationships between pairs of users. The exact features used by the model should be selected after checking their definitions and whether they contain meaningful variation.

3. Feature Analysis

3.1 Node-Level Features

The user dataset contains 36 columns. The supplied data include user and scenario identifiers, channel-related information, signal-to-noise information, slot-related information, and other features used by the project's scenario and labeling setup.

Examples of feature categories include:

- **Identifiers:** scenario and user identifiers used to associate records with their graph.
- **Channel and signal features:** channel gain and SNR-related values.
- **Spatial features:** user coordinates or distance-related information, where provided.
- **Assignment information:** slot and pairing fields used in the scenario setup.
- **Performance information:** rate- or constraint-related values used by the simulator.

Identifiers are generally needed for grouping and joining records, but they should not automatically be treated as predictive inputs. The final input feature set should be defined from the project schema and learning objective.

3.2 Edge-Level Features

The edge dataset contains 10 columns and 26,100 directed edge records. Edge records describe relationships between users within the scenario graph.

The edge table includes an oracle-pair indicator named `is_oracle_pair`. This label identifies the pairing decisions produced by the dataset's oracle labeling logic. It can be used as a supervised target if the intended learning task is to predict oracle pair relationships.

3.3 Scenario-Level Performance

The scenario summary contains aggregated information for each of the 290 scenarios. These records are useful for examining performance at the scenario level rather than treating each user or edge as an independent experiment.

Scenario-level performance measures should be evaluated using the same scenario grouping as the model's train, validation, and benchmark partitions.

4. Exploratory Statistical Analysis

4.1 Oracle Pair Label Distribution

The positive-label ratio for `is_oracle_pair` is approximately 0.1111, or 11.1%. Thus, positive edge labels form a minority of the edge records.

Table 4.1: Oracle edge-label distribution

Measure	Value
Positive-label ratio	11.11%
Negative-label ratio	88.89%

This imbalance affects the interpretation of classification results. A model that predicts the majority class can obtain a seemingly high accuracy while failing to identify useful positive pairs.

Therefore, accuracy should not be used alone. The evaluation should also report precision, recall, F1-score, and precision-recall AUC where appropriate. The choice of threshold should be determined using validation data, not the benchmark split.

4.2 Slot Distribution

Each scenario contains 10 users and considers five slots. The supplied dataset is constructed with balanced aggregate slot counts by design. This property describes the dataset construction and does not, by itself, demonstrate that a learned assignment method performs well.

When analysing slot distributions, the report should distinguish between aggregate balance across the dataset and the distribution within individual scenarios.

4.3 Scenario Performance Summary

The supplied scenario summaries give the following oracle sum-rate statistics.

Table 4.2: Oracle sum-rate summary by split

Split	**Mean (Mbps)**	**Standard Deviation (Mbps)**
Training	20.217	7.334
Validation	20.165	5.844
Benchmark	21.153	7.540

The benchmark split has a higher mean oracle sum rate than the training and validation splits in these summaries. This descriptive difference alone does not establish statistical significance or prove that a model generalizes better. It is important to report model results on the same benchmark scenarios and under a consistent evaluation protocol.

The average number of oracle outage users is reported as 0.52 for training, 0.60 for validation, and 0.58 for benchmark. These values describe the supplied scenario summaries and should be interpreted according to the definitions used by the dataset generator.

5. Correlation and Data Quality

5.1 Correlation Analysis

Correlation analysis helps identify features that contain similar information or have little variation.

The supplied data indicate that `snr_db` and `channel_gain_db` differ by a constant of 129 dB in magnitude under the dataset's stored convention. As a result, they have a perfect linear relationship in the supplied data. The precise interpretation of this relationship should be checked against the feature definitions and units before using both features in a model.

A feature named `edge_channel_correlation` has the constant value 1.0 for all supplied edges. A constant feature does not distinguish one edge from another and generally provides no predictive variation to a learning model.

Similarly, `node_min_rate_mbps` is constant at 0.1 for all supplied users. This feature should be reviewed to determine whether it is a fixed system constraint or an intended variable feature.

Table 5.1: Feature issues identified during EDA

Feature or Relationship	**Observation and Action**
`edge_channel_correlation`	Constant at 1.0. Confirm its intended meaning and consider excluding it from predictive inputs.
`node_min_rate_mbps`	Constant at 0.1. Confirm whether this is a fixed constraint rather than a varying user feature.
`snr_db` and `channel_gain_db`	Perfect linear relationship in the supplied dataset. Review units and definitions; avoid unnecessary duplicate inputs unless justified.
`is_oracle_pair`	Positive-label ratio is about 11.1%. Use metrics that account for class imbalance.

5.2 Basic Data Quality Checks

The reported checks on the supplied files found:

- No missing numeric values in the inspected tables.
- No duplicate `(scenario_id, user_id)` keys in the user table.
- No invalid partner-mapping symmetry cases in the checked partner mapping.
- The expected scenario and split counts are present.

These checks support the structural consistency of the supplied files. They do not prove that every feature is physically realistic, that the labels are optimal under every possible formulation, or that the dataset represents real wireless deployments.

6. Relevance to the Proposed GNN Model

6.1 Mapping the Dataset to a Graph Model

The supplied dataset is structurally relevant to a graph-based user-assignment and slot-pairing project.

Table 6.1: Dataset components and their potential role

Component	**Potential Role in the Project**
User records	Candidate node features for representing users.
Directed edge records	Relationships between users in a scenario graph.
Oracle pair labels	Supervised targets for predicting oracle pair relationships.
Slot information	Assignment-related information, subject to the chosen task definition.
Scenario summaries	Scenario-level performance analysis and evaluation.
Train/validation/benchmark splits	Separate model development, tuning, and benchmark evaluation.

6.2 Important Modeling Considerations

Before training, the project should define exactly what the GNN predicts. For example, edge classification can predict whether a candidate pair corresponds to an oracle pair. However, predicting an edge label is not automatically the same as solving the full user-assignment and slot-pairing optimization problem.

If the final objective is a valid assignment, the model output may require a separate decision procedure that enforces the project's constraints, such as the permitted number of users per slot or valid pairing rules.

6.3 Avoiding Data Leakage

The scenario identifier should be used to preserve scenario grouping and should not automatically be used as a predictive feature. Any preprocessing, normalization, feature selection, or threshold selection should be fitted using training data and, where needed, tuned using validation data.

The benchmark partition should remain separate until the evaluation stage. Users and edges from a single scenario must not be distributed across different partitions.

7. Limitations and Reproducibility

7.1 Synthetic Reconstruction

The dataset README describes the data as a reconstructed, synthetic dataset intended for the project's default GNN scenario setup. It reproduces configured distributions, seed indices, graph schema, and oracle-label logic.

The README also notes that the original project does not save the raw channel scenarios. The supplied reconstruction uses Python's standard-library `random.Random`, whereas the project code uses NumPy's `default_rng`. Therefore, the reconstructed files are not guaranteed to be the exact numerical samples generated by an earlier simulator run.

This provenance must be disclosed in the project report and presentation. If the project requires real measured wireless data, the supplied dataset alone does not meet that requirement. If a simulation-based evaluation is permitted, it can serve as a starting point, provided the reconstruction is documented.

7.2 Label Limitations

Oracle labels are generated from the dataset's oracle logic. They should be interpreted as targets produced by that procedure, rather than as independently measured ground truth from a real network.

Model performance against these labels measures agreement with the oracle labeling process. It does not, by itself, prove that the model improves performance in a deployed wireless system.

7.3 Packaging and Reproducibility

The dataset package should be checked before submission. The supplied README refers to an `EDA_Report.md` file that is not present in the described archive, and its example command uses a path that does not match the nested script location. Some figure filenames may also differ from the names referenced in the report.

For a reproducible submission, the project should:

- Include the actual EDA report in the dataset package.
- Verify the documented script path and execution command.
- Ensure all referenced figures exist and are readable.
- Document Python and library requirements.
- Record the exact feature columns used for training.
- Document the target label and scenario-level split policy.

8. Recommendations

Based on the exploratory analysis, the following actions are recommended before final model training and submission.

1. **Confirm the learning objective.** State whether the GNN predicts oracle edge labels, user assignments, slot assignments, or a combination of these.
2. **Review the feature set.** Investigate constant features and redundant signal-related variables. Keep only features justified by the task and their definitions.
3. **Account for label imbalance.** Report precision, recall, F1-score, and precision-recall AUC in addition to accuracy for edge classification.
4. **Preserve scenario-level separation.** Keep every user's records and associated edges within the same partition to reduce information leakage.
5. **Evaluate at the scenario level.** Include aggregate performance statistics across scenarios, not only metrics computed from individual edge records.
6. **Disclose dataset provenance.** Explain that the dataset is a synthetic reconstruction and state what that implies for reproducibility and conclusions.
7. **Complete the report package.** Verify figure names, script paths, execution instructions, and the inclusion of all supporting files.

9. Conclusion

The exploratory data analysis indicates that the supplied dataset has a consistent scenario-based structure suitable for developing a graph-learning workflow for user assignment and slot pairing. It contains 290 scenarios, 2,900 user records, and 26,100 directed edge records, with separate training, validation, and benchmark partitions.

The analysis also identifies important considerations for model development. These include the 11.1% positive oracle edge-label ratio, constant features, a perfect linear relationship between two signal-related features, and the need to preserve scenario-level split boundaries.

The dataset's synthetic reconstruction must be clearly disclosed. Oracle labels represent the dataset's labeling procedure and should not be equated with real-world ground truth. Consequently, this EDA supports the dataset's structural relevance to the proposed project but does not establish the performance of a trained GNN or validate real-world wireless-system improvements.

Future work should finalize the predictive task, review the input features, train and validate the model using scenario-level splits, and report benchmark results using metrics appropriate to the task and label distribution.

Data Sources

1. Supplied project dataset archive, `Dataset.zip`, including user records, edge records, scenario summaries, metadata, and the dataset README.
2. Supplied exploratory data analysis document, `ML_EDA_REPORT.pdf`.